Status	Finished
Started	Wednesday, 21 October 2026, 11:25 PM
Completed	Wednesday, 21 October 2026, 11:28 PM
Duration	2 mins 50 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. n ✓
- b. n/2
- c. log n
- d. n-1

Question **2**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- a. In the middle of the list
- b. In the right sublist ✓
- c. In the left sublist
- d. At the beginning of the list

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

- a. There must be a mechanism to delete and/or insert elements in the list ✓
- b. The list must be sorted
- c. There should be direct access to the middle element in any sublist
- d. Number values should only be present

Number values should only be present

Question **4**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

- a. O(1) ✓
- b. O(log n)
- c. O(n log n)
- d. O👎

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

- a. Sorting the list before searching
- b. Checking each element sequentially ✓
- c. Skipping every second element
- d. Dividing the list in half repeatedly

Question **6**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

- a. Binary ✓
- b. Both (1) & (3)
- c. Hash
- d. Linear

Question **7**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the element is found in the list ✓
- b. When the search algorithm finishes
- c. When the list contains duplicate elements
- d. When the list is sorted

Question **8**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

- a. By comparing each element in the list sequentially ✓
- b. By dividing the list into halves
- c. By using a hash function
- d. By sorting the list first

Question **9**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

- a. Mining
- b. Finding
- c. Searching ✓
- d. Discovering

Question **10**

Incorrect

Mark 0.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

- a. 89 and 99
- b. 89 and 94
- c. 90 and 94 ✗
- d. 90 and 99

Question **11**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

- a. Linear search
- b. Sequential search
- c. Hash search
- d. Binary search ✓

Question **12**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search continues from the beginning
b. The list is sorted
c. The search stops
d. The search continues in the left or right sublist ✓

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It can only be applied to large lists
b. It does not work with negative numbers
c. It is slower than linear search for small lists
d. It requires the list to be sorted ✓

Question **14**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Binary search
b. Linear search ✓
c. Hash search
d. Both (1) & (3)

Question **15**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It can be applied only if the list is sorted ✓
b. It compares elements sequentially
c. It works on unsorted lists
d. It always starts from the beginning of the list